

Fusion Science and Technology

ISSN: 1536-1055 (Print) 1943-7641 (Online) Journal homepage: http://www.tandfonline.com/loi/ufst20

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To cite this article: Alexander I. Livshits, Yuji Hatano & Kuniaki Watanabe (2002) Superpermeability in Fusion Technology: Tritium Accumulation and Compression, Fusion Science and Technology, 41:3P2, 882-886, DOI: 10.13182/FST02-A22711

To link to this article: http://dx.doi.org/10.13182/FST02-A22711

Published online: 10 Aug 2017.

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SUPERPERMEABILITY IN FUSION TECHNOLOGY: TRITIUM ACCUMULATION AND COMPRESSION

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ABSTRACT

Superpermeable membranes based on Group Va metals can be applied in fusion devices for a short way separation of D/T mixtures from He, for an active control of particle fluxes and as a general-purpose D/T pump that may be used in particularly in tritium handling systems. Superpermeable membranes being used for D/T separation from helium are able to drastically reduce the tritium load on the He pump (cryopump), while tritium accumulation in the membrane itself does not exceed a few g for a machine of ITER scale. A possible way to decrease the tritium inventory in the membrane is to combine a higher dissociative barrier at the upstream surface with the operation at higher temperature. Compression of permeating D/T attainable with superpermeable membranes is totally determined by the sticking coefficient of thermal hydrogen molecules at the upstream surface. The degree of compression has a significant effect on the tritium inventory and the inventory dependence on the state of the downstream surface.

I. INTRODUCTION

A metallic membrane of macroscopic thickness may be superpermeable to hydrogen particles whose energy (kinetic, internal or chemical) exceeds ~ 1 eV [1]. That means that virtually the whole flux of suprathermal particles implanted in the metal lattice, or just sticking to its surface, passes through the membrane, with the permeation flux depending neither on membrane temperature nor on its thickness. Since the probability of implantation (or of sticking) of suprathermal particles is comparable or even close to unity and does not depend on the metal temperature, the permeation probability also is comparable/close to unity and temperature independent at superpermeability.

Superpermeable membranes (SPM) have a number of merits for their applications in fusion devices. They are (1) the maximum conceivable rate of suprathermal hydrogen evacuation with a possibility to operate at flux densities as high as ~10^{19} $cm^{-2}s^{-1}$, (2) the steady state operation, (3) the extremely high selectivity in permeation of hydrogen (tritium) among other gases including He, (4) the automatic compression of permeating hydrogen by orders of magnitude, (5) the low hydrogen (tritium) accumulation, (6) absence of any mechanically moving parts, (7) the ability to operate in a wide temperature range (from 20 to ~1000 °C) (8) a possibility to operate in a strong magnetic field. Points (1)-(3), (5), (6) appear to be especially important for tritium operation.

Short way separation of D:T mixtures from He is one of the proposals putting forward on SPM applications in fusion [2]. Helium is supposed to be pumped by cryopumps in the future fusion projects (e.g. in ITER). SPM are proposed to be installed along the pumping duct walls before the cryopumps. If SPM are installed far enough from the divertor, they should have an own generator of suprathermal particles, e.g. a filament atomizer or a gas discharge. As it has been shown [2], on the basis of the results of model experiments, a membrane system of quite a reasonable size and with a fairly moderate power consumption can isolate more than 99% of D/T, drastically reducing its freezing at the cryogenic panels.

Another proposal is connected with an idea that an effective hydrogen pumping from the divertor region is expected to result in an enhancement of the core plasma temperature [3]. For example, the idea of pumping consists of the removal with a membrane panel of the charge-exchange atoms generated by divertor plasma. Such a membrane panel is proposed [3] for non-tritium devices but would be especially effective in the case of actual tritium operation.

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In addition, superpermeable membranes can serve as a general-purpose hydrogen/tritium pump in any auxiliary systems of a fusion device including the tritium handling system.

This work is focused on the points specifically important for the tritium applications but never specially discussed before: the accumulation of tritium in the SPM and the D/T mixture compression by the SPM that is closely linked with the accumulation.

II. PHYSICOCHEMICAL CAUSES OF THE PHENOMENON

Potential barriers for the processes of H_2 dissociative dissolution / associative release (E_{ru} and E_{rd} in Fig. 1) are generated at a surface of transition metals by nonmetallic impurity monolayers [1, 4, 5], typically remaining at a metal surface after a moderate heating in vacuum. Suprathermal hydrogen particles with an excess of energy (kinetic, internal, or chemical) are able to freely penetrate to the inside of the metal in spite of the barrier. Hydrogen atoms "cooled down" in metal lattice can not be easily released back through the barrier $E_{ru} - E_b$ (Fig. 1). If the "reflection" from this barrier is strong enough (the "strong reflection condition" [1]), absorbed atoms can reach the membrane downstream side by diffusion easier than be released back. The absorbed atoms are reflected in the same way from the downstream barrier, $E_{rd} - E_b$ (Fig. 1), and thus they are crossing the membrane many times and "forget" their initial position (with the concentration of solute hydrogen nearly leveling off across the whole membrane thickness). As a result, hydrogen is desorbed to the left or to the right depending only on relative surface barrier heights. If $E_{ru} > E_{rd}$, a major part of implanted particles will pass through the membrane regardless of membrane temperature and thickness.

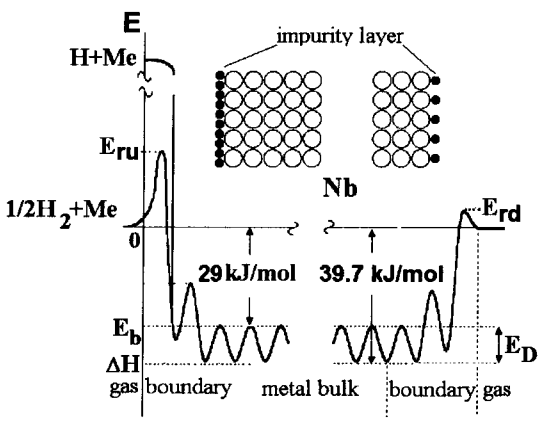


Fig. 1. Potential energy diagram for a hydrogen-niobium membrane system. The energy of an H atom in H_2 molecule is taken for the zero energy level.

Group Va metals (V, Nb and Ta) appear to be most suitable as materials for SPM. This is due to a favorable combination of three factors: the lowest level of barrier E_b among other metals (specifically for these metals $E_b < 0$, Fig. 1 [6]), the very low diffusion barrier E_D [6, 7], and the high chemical reactivity resulting in an existence of thermally and chemically stable nonmetal monolayers (e.g. of oxygen) at the surface generating a high barrier of hydrogen dissociation/association E_{ru} [5, 8]. As a result the strong reflection regime [1] takes place over a wide range of flux densities (up to ~10^{19} $cm^{-2}s^{-1}$), temperatures (up to ~1000 °C), and thickness (up to 1 cm). Correspondingly superpermeation through Group Va metals appears over this range, if the symmetry of hydrogen release stays favorable (i.e. $E_{ru} > E_{rd}$).

III. COMPRESSION AND TRITIUM ACCUMULATION

A. Compression of Permeating D/T

At temperatures lower than ~1000 °C, absorbed hydrogen can be desorbed from the downstream membrane surface only in the form of molecules. However, the membrane superpermeable to suprathermal hydrogen exhibits a low permeability to thermal molecules. Therefore, permeating gas can not easily penetrate back and may be automatically compressed. If there is no pumping at the downstream side, downstream pressure, p_d, increases up to a value of p_{dmax} corresponding to a flux balance between the permeation of molecules from the downstream side and that of suprathermal hydrogen from the upstream side. At the balance, gas at the downstream side is in equilibrium with absorbed hydrogen:

$$p_{d\,max} = c_b^2 / S^2 . \quad (1)$$

Here, S is the solubility constant and c_b is the concentration of absorbed hydrogen at the balance that may be found from the flux balance equation:

$$j_i \xi = k_{ru} c_b^2 , \quad (2)$$

where j_i is the density of suprathermal hydrogen flux, ξ is sticking/implantation coefficient for suprathermal particles and k_{ru} is the rate constant of thermal release through the upstream surface. With using the detailed balance principle, k_{ru} may be expressed through the probability of molecule absorption (sticking coefficient) α_u as [5]

$$k_{ru} = z\alpha_u / S^2 , \quad (3)$$

where z is the gas kinetic theory coefficient. From Eqs. (1) - (3)

$$p_{d\,max} = \frac{j_i \xi}{z \alpha_u} . \quad (4)$$

One should note that the maximum compression pressure $p_{d\,max}$ depends *only* on the implanted flux density and properties of upstream surface, i.e. on α_u (without any dependence on membrane material and thickness, state of downstream surface, regime of permeation, *etc*). Next, α_u can be presented as

$$\alpha_u = \alpha_{u0} \exp(-\frac{2E_{ru}}{RT}) , \quad (5)$$

where $2E_{ru}$ is the barrier of dissociative absorption of hydrogen molecule (Fig. 1). With Eq. (5),

$$p_{d\,max} = \frac{j_i \xi}{z \alpha_{u0}} \exp \frac{2E_{ru}}{RT} . \quad (6)$$

Thus the stronger the passivation of upstream surface (i.e. the higher E_{ru}, the lower α_{u0}), the higher the attainable compression. As it was recently found [8, 9], the values of E_{ru} and α_{u0} may be controlled by doping of nonmetal impurities in the metal bulk. For instance, doping Nb with 1.5 atomic % of O results in $\alpha_u = 5.5 \cdot 10^{-4} \exp[-41.8$ (kJ/mol)/RT]. Fig. 2 gives $p_{d\,max}$ estimated with this value of α_u and $j_i \xi \approx 1 \cdot 10^{16}$(D/T)/(cm^{2}s) (one will have such a flux density, if SPM is used to separate D/T from He in ITER [2]).

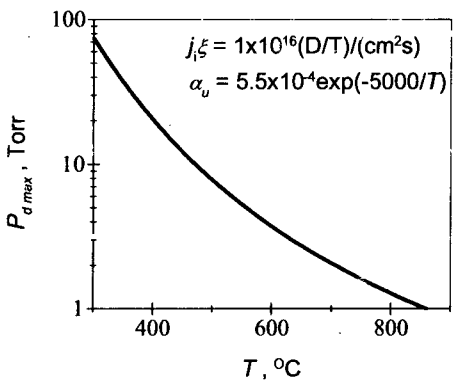


Fig. 2. Temperature dependence of maximum downstream pressure (compression), which Nb membrane doped with O may provide.

However, the net permeation flux is zero at $p_d = p_{d\,max}$. So p_d should be kept well lower than $p_{d\,max}$ at actual operation, e.g. $p_d \le \varepsilon\, p_{d\,max}$, where, for instance, $\varepsilon \approx 0.1$.

B. Steady-State Hydrogen Concentration

According to the above, the strong reflection regime always takes place in the case of membrane of Group Va metals. This means that hydrogen concentration c is virtually constant across the membrane and c may be found at the steady state permeation with compression of permeating gas from

$$j_i \xi + p_d z \alpha_d = c^2 (k_{ru} + k_{rd}) \quad \text{at} \quad p_d \le p_{d\,max} ,$$

where α_d is probability of molecule absorption through the downstream surface, k_{ru} and k_{rd} are the rate constants of hydrogen release through the up- and downstream surfaces, respectively. With Eq. (3) and a similar equation for k_{rd}, it follows:

$$c = (\frac{j_i \xi + p_d z \alpha_d}{\alpha_u + \alpha_d} \frac{S^2}{z})^{\frac{1}{2}} \quad \text{at} \quad p_d \le p_{d\,max} . \quad (7)$$

The condition of preferable release of absorbed hydrogen through the downstream side is $\alpha_d >> \alpha_u$. This favorable membrane asymmetry can be realized practically, e.g. by formation of a certain roughness at the downstream surface. For such asymmetrical membrane c can be expressed as:

$$c \approx (\frac{j_i \xi}{z \alpha_d} + p_d)^{\frac{1}{2}} S \quad \text{at} \quad p_d \le p_{d\,max} . \quad (8)$$

Next, α_d is given similar to α_u (Eq. (5), Fig. 1)

$$\alpha_d = \alpha_{d0} \exp(-\frac{2E_{rd}}{RT}) , \quad (9)$$

and

$$S = S_0 \exp(-\frac{\Delta H}{RT}) = S_0 \exp \frac{|\Delta H|}{RT} , \quad (10)$$

where ΔH is the solubility enthalpy. ΔH is negative for Group Va metals (Fig. 1) and we use module sign to emphasize that exponent in Eq. (10) is positive. With Eqs. (8)-(10),

$$c \approx \sqrt{(\frac{j_i \xi}{z \alpha_{d0}} \exp \frac{2E_{rd}}{RT} + p_d) S_0 \exp \frac{|\Delta H|}{RT}} \quad \text{at} \quad p_d \le p_{d\,max} . \quad (11)$$

According to Eq. (11), the higher T, the lower c. The operation at a high temperature can be easily arranged (e.g., with using the redundant power from filament or plasma atomizer of molecular hydrogen [10] wasted on radiation). But, according to Eq. (6), the required p_d (compression) cannot be reached, if the temperature is too high. A maximum temperature compatible with the required p_d can be found by Eq. (6) with $p_{d\,max} = p_d / \varepsilon$:

$$T_{\max}(p_d) = \frac{2E_{ru}}{R} \frac{1}{\ln[p_d \alpha_{u0} z / (j_i \xi \varepsilon)]} . \quad (12)$$

Substituting that into Eq. (11), we come to a minimum hydrogen concentration in the metal at a given p_d (compression): $c_{min}(p_d)$. If the downstream surface differs from the upstream one only by its greater roughness (made specially for the purpose of a favorable asymmetry), then $E_{rd} = E_{ru}$, the asymmetry being due to a difference in the pre-exponential factors: $\alpha_{d0} >> \alpha_{u0}$. In this case,

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$$c_{\min}(p_d) = [p_d(\frac{\alpha_{u0}}{\alpha_{d0}\varepsilon} + 1)]^{\frac{1}{2}} S_0 (\frac{p_d \alpha_{u0} z}{j_i \xi \varepsilon})^{\frac{|\Delta H|}{2E_{ru}}} . \quad (13)$$

According to Eq. (13), the hydrogen accumulation may be decreased by higher upstream barrier E_{ru} (Fig. 1) and lower pre-exponent α_{u0} (Eq. (5)). Thus a possible way to reduce the hydrogen accumulation with keeping the required compression is combining a stronger passivation of the upstream surface with the operation at higher temperature (Eq. 12). Fig. 3 demonstrates that for two values of $2E_{ru}$: the experimentally observed for Nb doped with O (41.8 kJ/mol) and the higher one by 1.5 times.

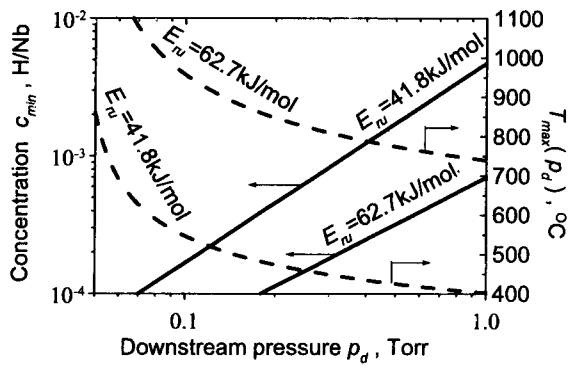


Fig. 3. Dependence of relative hydrogen concentration in Nb membrane on the downstream pressure (compression) at the maximum membrane temperature compatible with this pressure (this temperature is also plotted here). The value of dissociative barrier E_{ru} is a parameter of the plot. The calculation was made by Eqs. (12), (13) taking $j_i\xi = 2\cdot10^{16}$ H/(cm^2 s), $\alpha_{u0} = 5.5\cdot10^{-4}$, $\varepsilon = 0.1$, $\alpha_{d0} = 10\alpha_{u0}$, $S_0 = 0.6\cdot10^{18}$ H/(cm^3 Torr$^{1/2}$), $\Delta H = -39.7$ kJ/mol.

Another conclusion as regards the influence of the state of downstream surface on hydrogen accumulation follows from Eqs. (11) and (13). If compression is not produced (i.e., if p_d in Eq. (11) is always negligible), any activation of the downstream surface (a decrease of E_{rd} or increase of α_{d0}) decreases c (Eq. (11)). But if the permeating gas must be compressed, making the downstream surface very active would be of no use: the effect of activation has saturation. For instance, according to Eq. (13), such a saturation commences at $\alpha_{d0} \geq \alpha_{u0}/\varepsilon$. Hence on taking $\varepsilon \approx 0.1$ we come to $\alpha_{d0}/\alpha_{u0} \approx 10$, as a reasonable membrane asymmetry.

C. Tritium Inventory

Let us estimate as an example the tritium inventory G in the SPM of Nb, if it would be used for D/T separation from He in a fusion device of ITER scale (section I). $G = cBL$, where L is the membrane thickness and B is the membrane area that can be presented as $B = F_m/f_m$ where F_m is the required speed of membrane pumping and f_m is the pumping speed per unit of membrane area. If tritium is getting with the speed F_{He} into the cryopump designed to pump He, and, if one wants to reduce the tritium load on the cryopump due to SPM by a factor of A, one should have $F_m = (A-1)F_{He}$. With $A >> 1$, the density of D/T flux absorbed by the membrane is expressed as $j_i\xi \approx J/B$, where J is the total D/T influx into the fusion device. With taking that into account and with using Eq. (7), one gets:

$$G = \sqrt{\frac{AF_{He}/f_m[J/(\alpha_d z) + p_d AF_{He}/f_m]}{1+\alpha_u/\alpha_d}} SLm \,, (14)$$

where m is tritium atom mass. For Nb membrane combined with a filament atomizer [2], the experimentally obtained specific pumping speed (with protium) is $f_m \approx 1.5$ l/(cm^{2}s). Taking $J \approx 10^{22}$ (D/T)/s, and $F_{He} \approx 10^5$ l/s as typical values for a machine of the ITER scale [11], and using $A = 10$, $L = 0.01$ cm and $p_d = 0.1$ Torr, we calculated G (T) at several values of the membrane asymmetry, α_d/α_u (Fig. 4). In this estimation, we used for tritium the same value of α_u that was taken above for protium: $\alpha_u = 5.5\cdot10^{-4}\exp-[41.8\text{ (kJ/mol)}/RT]$, assuming as before α_d to differ from α_u only by pre-exponential factor. We also used for protium solubility constant $S = 0.6\cdot10^{18}\exp[39.7\text{(kJ/mol)}/RT]$ H/(cm^3 Torr$^{1/2}$) [6]. The solubility constant for tritium in Nb is unknown. But usually the isotope effect in hydrogen solubility is rather small (e.g., it is practically negligible for the solubility of H and D in vanadium [12]).

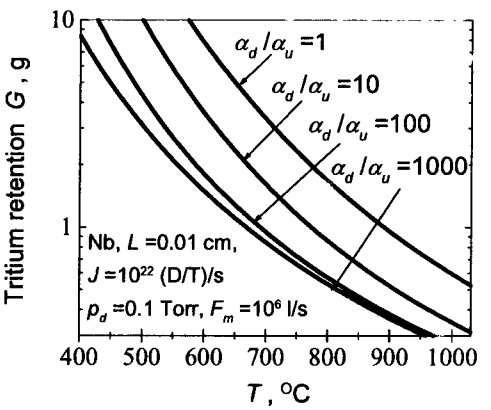


Fig. 4. Temperature dependence of tritium inventory in the Nb membrane separating ~90% D/T from He before a He pump in a fusion device of ITER scale (calculated by Eq. (14)).

Effect of temprature The strong temperature dependence of G demonstrated in Fig. 4 originates mainly from a large enthalpy of hydrogen solution in Nb, ΔH (Fig. 1). If $T \geq 600$ °C, *the tritium inventory is not expected to exceed a few grams* in our example, but small T increase

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should result in a desirable G decrease anyway. Such temperatures can be easiliy attained in practice. For example, the membrane temperature was automatically keeping at a level of 700-750 °C in the laboratory-scale experiments that modeled the real configuration and all the specific (per unit area) characteristcs of the membrane system with a filament atomizer proposed for the separation D/T from He in fusion (section 1, [2]). The membrane was heated by radiation from the atomizer filaments [10]. A simple thermal radiation shield increased this temperature to ~1000 °C.

If it is necessary to operate at a lower temperature, the membrane should be made of materials with a lower heat of solution ΔH, e.g. of vanadium based alloys.

Effect of asymmetry In accordance with the description in section B, favorable membrane asymmetry noticably reduces the tritium accumulation in the considered example (Fig. 4) though compressing the permeating gas (up to 0.1 Torr) weakens the possible asymmetry effect (especially at higher temperatures) and results in its saturation.

D. Superpermeation with Vanadium Based Membranes

Practically, V or V-based alloys should be used for SPM in actual tritium machines rather than Nb because of their low induced radioactivity by neutron irradiation. All bulk properties of V-hydrogen system are very close to those in Nb-hydrogen system [6, 12]. But the boundary kinetics and effects of nonmetallic impurities (e.g. O) on it, playing the key role, are scarcely investigated for vanadium-hydrogen system. Thus studies of boundary kinetics and superpermeration with V based membranes including tritium experiments are interesting and practically important task.

IV. CONCLUSIONS

SPM based on Group Va metals may be applied in fusion devices for short way separation of D/T from He, for active control of particle fluxes and for a general D/T pumping in tritium handling system. Compression of permeating D/T with SPM is totally determined by the sticking coefficient of thermal hydrogen molecules to the upstream surface and can be controlled with concentration of nonmetal impurities in the metal bulk. SPM being used for D/T separation from He are able to drastically reduce the tritium load on the He pump (cryopump) while the tritium accumulation in the membrane is expected not to exceed a few g for the machine of ITER scale. Degree of compression has a substantial effect on the tritium inventory and its dependence on membrane asymmetry. A possible way to decrease the tritium inventory in the membrane is to combine a stronger passivation of upstream surface with operation at higher temperature.

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